

PX-200 Industrial Circulation Pump

Operating and Maintenance Manual - Revision 2.4

This is a fictional document created for AI testing training. The product, values and procedures are invented. Do not use for real equipment.

1. General safety notes

- Read this manual completely before installing, operating or servicing the pump.
- Only trained personnel may carry out installation and maintenance work.
- Wear safety glasses, gloves and hearing protection when working near the pump in operation.
- Keep the area around the pump clear of tools and loose material.

2. Product overview

The PX-200 is a heavy-duty centrifugal pump for higher-pressure circulation in large production lines. It shares its housing design with the PX-100 but is rated for higher pressure and flow.

3. PX-200 operating limits

Operate the PX-200 only within the limits below. Operating outside these limits can damage the pump and void any service agreement.

Parameter	Limit
Maximum operating pressure	15 bar
Fluid temperature range	15 - 45 °C
Maximum flow rate	80 m3/h
Motor power	11 kW
Supply voltage	400 V, 3-phase, 50 Hz

Do not bypass or disable the pressure interlock under any circumstances.

4. Installation

- Mount the pump on a level, vibration-free base.
- Align the pump and motor shafts before first start-up.
- Ensure at least 30 cm of free space around the motor for ventilation.
- Fill the pump casing with fluid before starting. Never run the pump dry.

5. Preventive maintenance

Preventive maintenance for the PX-200 is required every 3 months. Record each maintenance visit in the equipment log.

Task	Interval
Inspect seals for leakage	3 months
Check bearing temperature and noise	3 months
Replace lubricant	Every 6 months
Check shaft alignment	Every 12 months

6. Preparing the pump for maintenance

1. Switch off the pump and isolate the electrical supply at the main switch.
2. Apply lock-out/tag-out to the main switch so the pump cannot be restarted.
3. Close the inlet and outlet valves.
4. Release the pressure from the pump casing using the drain valve, and confirm the gauge reads 0 bar.
5. Allow the pump to cool to below 40 °C before touching the casing.
6. Only then open the casing or inspect the impeller. Never inspect the impeller while the pump is running.

7. Troubleshooting: overheating

Possible cause	Action
Blocked cooling path or poor ventilation	Clear obstructions; check 30 cm clearance
Low or degraded lubricant	Replace lubricant (see section 5)
Running above rated pressure	Reduce load; do not exceed 15 bar
Worn bearings	Inspect bearings; replace if noisy or hot
Shaft misalignment	Re-align pump and motor shafts

If the pump overheats repeatedly, stop operation and raise a service request with your maintenance team.

8. Contact

For service requests, contact your site maintenance team using the internal service request process.